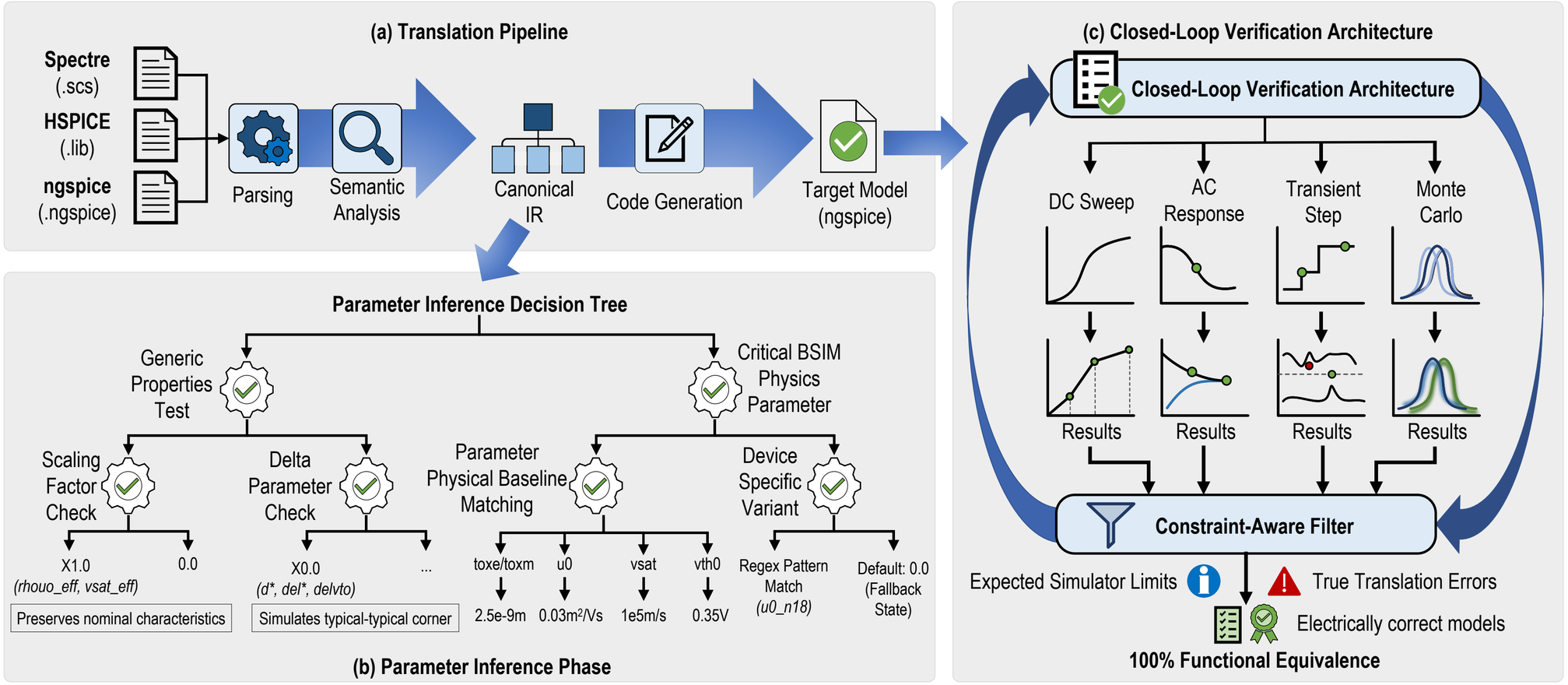

(a) Translation Pipeline
Spectre
(.scs)
HSPICE
(.lib)
ngspice
(.ngspice)
Parsing
Semantic
Analysis
Canonical
IR
Code Generation
Target Model
(ngspice)
Parameter Inference Decision Tree
Generic
Properties
Test
Critical BSIM
Physics
Parameter
Scaling
Factor
Check
Delta
Parameter
Check
Parameter
Physical Baseline
Matching
Device
Specific
Variant
X1.0
0.0
(rhouo_eff, vsat_eff)
X0.0
...
(d*, del*, delvto)
toxe/toxm
u0
vsat
vth0
Regex Pattern
Match
(u0_n18)
Default: 0.0
(Fallback
State)
Preserves nominal characteristics
Simulates typical-typical corner
2.5e-9m
0.03m²/Vs
1e5m/s
0.35V
(b) Parameter Inference Phase
(c) Closed-Loop Verification Architecture
Closed-Loop Verification Architecture
DC Sweep
AC
Response
Transient
Step
Monte
Carlo
Results
Results
Results
Results
Constraint-Aware Filter
Expected Simulator Limits
True Translation Errors
Electrically correct models
100% Functional Equivalence